

Impact of RIN Transfer and Pump-Induced XPM on Coherent Transmission with Forward-Pumped-Raman Amplification

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Abstract We propose a numerical framework for estimating the impact of pump-to-signal RIN transfer and pump-induced XPM on QoT under forward-Raman pumping. Numerical estimates show reasonable agreement with experiments in single-channel and WDM configurations. The results indicate that pump-induced XPM is more significant than RIN transfer. ©2026 The Author(s)

Introduction

Distributed-Raman amplification (DRA) is a promising technique for improving optical transmission performance [1]. In contrast to backward DRA, forward DRA (FDRA) is less commonly used because co-propagating Raman pumps and wavelength-division multiplexing (WDM) channels can induce signal impairments. A representative example is pump-to-signal relative-intensity-noise (RIN) transfer through stimulated Raman scattering (SRS) [2]. Nevertheless, FDRA offers a clear advantage in reducing the maximum power within the span via SRS gain, thereby mitigating nonlinear impairments such as self-phase modulation (SPM) and inter-channel cross-phase modulation (XPM) [3]. FDRA is also expected to be beneficial for multiband transmission, including additional bands such as the S-band [4], that suffers from inter-channel SRS and the lower saturation output power of lumped amplifiers. In addition, semiconductor incoherent light sources (iPUMPs) [5][6], broadband pump sources [7], and a longitudinal-mode-optimized pumping scheme [8] have been developed to mitigate the impact of RIN transfer in FDRA.

In intensity modulation and direct detection systems, RIN transfer has been extensively studied, including the effects of dual-order pumping [9], pump bandwidth [10], and polarization [11]. In digital coherent systems, the impact of pump RIN has also been investigated in terms of pump-RIN-induced signal relative phase noise [12]. However, these effects have been treated separately, so a framework suitable for the design of coherent systems using FDRA is still lacking.

In this study, we propose a numerical framework for estimating the quality of transmission (QoT) of digital coherent systems using FDRA while accounting for pump-induced impairments. The framework incorporates pump-to-signal RIN transfer and pump-induced XPM into a Gaussian noise (GN) model-based QoT estimation scheme [13]. The numerical estimates are experimentally validated through single-channel and C-band WDM transmission.

Proposed numerical framework

This section outlines the proposed numerical framework, whose conceptual diagram is shown in Fig. 1. In addition to the conventional QoT penalties, including transceiver noise, nonlinear interference (NLI) associated with SPM and inter-channel XPM and amplified spontaneous emission (ASE) from lumped amplifiers, we introduce forward-Raman-pump-induced impairments, highlighted in red. Specifically, pump-to-signal RIN transfer and pump-induced XPM are considered. Both are approximately treated as equivalent Gaussian noise terms, similarly to SPM- and inter-channel-XPM-induced NLI estimated by the GN model [14]. This approximation is justified by the significant temporal spreading of WDM signals caused by chromatic dispersion, making the Gaussian approximation reasonable, as in [15].

Pump-to-signal RIN transfer is modelled by linearizing coupled Raman equations with pump-signal walk-off and expressing the modulation-index evolution at RIN angular frequency Ω in a state-transition-matrix form $\mathbf{H}(\Omega; z)$. $\mathbf{H}(\Omega; z)$ describes a transition from the fibre input to position z and is obtained by solving

$$d\mathbf{H}(z, \Omega)/dz = \mathbf{A}(z, \Omega)\mathbf{H}(z, \Omega), \mathbf{H}(0, \Omega) = \mathbf{I} \quad (1)$$

where the diagonal terms of $\mathbf{A}(z, \Omega)$ represent walk-off and the off-diagonal terms represent Raman coupling determined by the frequency-dependent average powers and Raman gain coefficients. Raman pumps are represented by frequency-bin discretization and a stochastic input covariance matrix $\Sigma_{in}(\Omega)$, derived from the measured pump optical spectrum, RIN spectrum, and an inter-bin correlation parameter $\eta \in [0, 1]$. Here, $\eta = 0$ corresponds to independent RIN among bins, whereas $\eta = 1$ corresponds to complete correlation. The transferred signal RIN is

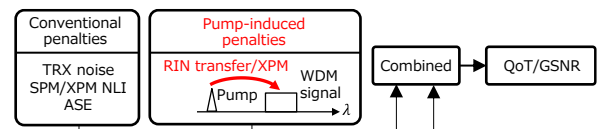


Fig. 1: Conceptual diagram of proposed QoT estimation framework for forward-Raman-amplified coherent transmission.

obtained from the output covariance matrix $\Sigma_{out}(\Omega) = \mathbf{H}(\Omega; z) \Sigma_{in}(\Omega) \mathbf{H}(\Omega; z)^H$, and converted into a QoT penalty by in-band integration.

To estimate pump-induced XPM, we treat the forward Raman pump as a nonlinear interfering field acting on the signal through the Kerr effect. Since the pump does not generally satisfy the signal-Gaussianity assumption underlying the GN model, the effect of the correlation among its spectral components must be considered. Furthermore, forward Raman pumps are launched into each fibre span, so chromatic dispersion accumulates only within a single span for the pumps, unlike WDM signals, which experience dispersion accumulation over multiple spans. As a result, the pump field is not expected to be well approximated as Gaussian noise even in long-haul transmission. We therefore use the enhanced GN (EGN) model [16], in which the non-Gaussian pump statistics are represented through the fourth-order moment μ of the pump complex field, while assuming a zero-mean circular distribution. In EGN numerical integration, SRS-induced power transfer is taken into account by using the link function expression in [17]. The limitation of this approach originates from the first-order perturbation approximation [18], which may lead to overestimation of pump-induced XPM when XPM becomes too strong [19].

The proposed RIN model is extended to multiple frequency components so that it can be applied to arbitrary numbers of signal channels and pumps. It is also formulated in a stochastic manner to maintain consistency with the EGN-based pump-induced XPM model. In addition, the role of signal modulation-format dependence in the conventional EGN framework is reinterpreted in terms of pump-field statistics, allowing the non-Gaussian characteristics of the pumps to be incorporated through their fourth-order moment.

Experimental validation and discussion

Fig. 2 shows the experimental setup for recirculating single-channel and WDM transmission under forward Raman amplification. For the WDM experiments, 45 channels with 100-GHz spacing were emulated using ASE spanning 1528.2–1564.0 nm as shown in Fig. 3(a). A 96-Gbaud polarization multiplexed probabilistically constellation-shaped (PCS)-36QAM signal was used as the channel under test (CUT), which was

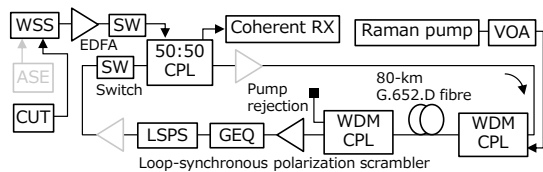


Fig. 2: Experimental setup for recirculating single-channel and WDM transmission under forward Raman amplification.

multiplexed by a wavelength selective switch (WSS). The recirculating loop comprised an 80-km G.652.D fibre, WDM couplers (CPLs) for multiplexing and demultiplexing Raman pumps, a gain equalizer (GEQ), and other optical components for recirculating transmission. At the receiver, CUT was demodulated offline using digital signal processing with a complex 8×2 MIMO equalizer [20] with a data-aided algorithm. The constellation SNR was evaluated from the recovered symbols and compared with the calculated SNR; the conventional calculation excluding pump-induced penalties is described in [21]. For the single-channel experiments, the gray-shaded components in Fig. 2 were removed, and CUT was set to 1530 nm as indicated in Fig. 3(a). Fig. 3(b) shows the optical spectra of forward Raman pumps. Pump powers were normalized to compare spectral shapes. Fiber-Bragg-grating-laser-diode-based coherent pumps (cPUMPs) at 1423, 1480 and 1505 nm, as well as an iPUMP centered at 1423 nm, were evaluated. The maximum launch powers of polarization-multiplexed pumps into the fibre were 24.4, 24.6, 23.3, and 22.6 dBm for the 1423-, 1480-, and 1505-nm cPUMPs and the 1423-nm iPUMP, respectively. The pump current settings were kept fixed at the maximum-power conditions, and the launch power was adjusted using a variable optical attenuator (VOA).

Fig. 4(a) shows the pump-to-signal RIN transfer function estimated with the proposed RIN model for the 80-km single-channel transmission. Representative cases for the 1423-nm cPUMP with 10.0 dB Raman gain and the 1480-nm cPUMP with 5.2 dB gain are presented. The cut-off frequency slightly increased for the longer-wavelength pump because of the smaller pump-signal walk-off, which is consistent with previous studies. Figs. 4(b)–(d) show measured pump RIN spectra at the input and the calculated signal RIN spectra after 80-km transmission for the 1423- and 1480-nm cPUMPs and the 1423-nm iPUMP. The calculated signal RIN spectra were consistent with both the estimated RIN transfer functions and the measured pump RIN spectra.

Fig. 5(a) shows measured and calculated SNR as a function of on-off forward Raman gain after 1280-km single-channel transmission at a launch power of 1 dBm/ch. Results for the 1423-, 1480-, and 1505-nm cPUMPs and the iPUMP are

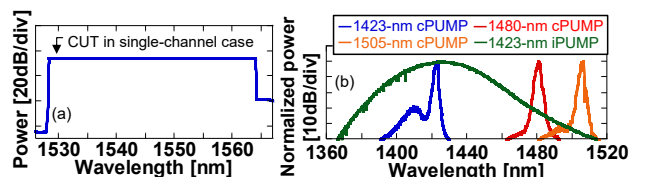


Fig. 3: Optical spectra of (a) ASE-emulated WDM signals and (b) forward Raman pumps.

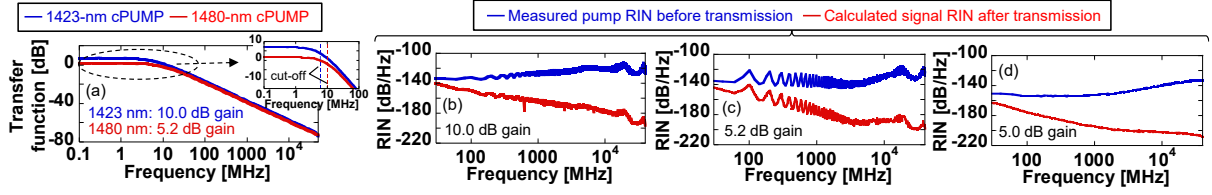


Fig. 4: RIN transfer estimation. (a) Calculated RIN transfer functions for cPUMPs. Measured pump RIN spectra at fibre input and calculated signal RIN spectra after 80-km transmission for (b) 1423- and (c) 1480-nm cPUMPs, and (d) 1423-nm iPUMP.

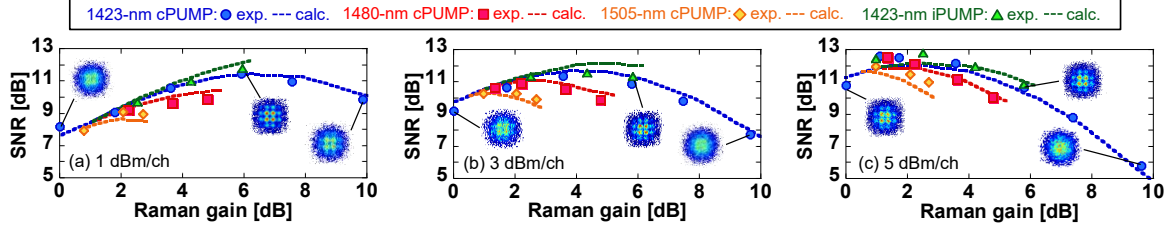


Fig. 5: Measured and calculated SNR as function of on-off forward Raman gain for signal launch powers of (a) 1 dBm/ch, (b) 3 dBm/ch, and (c) 5 dBm/ch in 1280-km single-channel transmission.

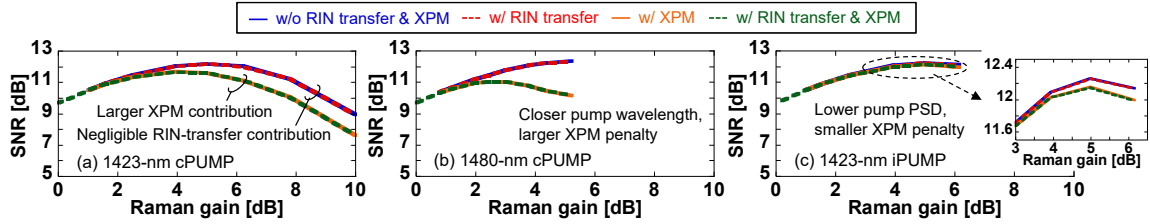


Fig. 6: Calculated SNR for comparing impacts of RIN transfer and pump-induced XPM in 1280-km single-channel transmission at 3 dBm/ch for (a) 1423-nm cPUMP, (b) 1480-nm cPUMP, (c) 1423-nm iPUMP.

presented. In the calculation, the inter-bin correlation parameter η and fourth-order moment μ were chosen to provide reasonable agreement with the measured results. The resulting parameters were $\eta = 1$, $\mu = 1.7$ for the 1423-nm cPUMP; $\eta = 1$, $\mu = 1$ for the 1480-nm cPUMP; $\eta = 1$, $\mu = 1$ for the 1505-nm cPUMP; $\eta = 1$, $\mu = 2$ for the 1423-nm iPUMP. Note that the calculated SNR was insensitive to η , and $\eta = 1$ corresponds to the worst-case assumption. Fig. 5(b) and (c) show the same comparison at 3 and 5 dBm/ch, respectively, using the same η and μ . The calculated SNR was generally consistent with the measured SNR, supporting the validity of the proposed framework.

Fig. 6(a) compares the impacts of the RIN transfer and pump-induced XPM for 1280-km single-channel transmission at 3 dBm/ch with the 1423-nm cPUMP. The calculated SNRs without either impairment and with only RIN transfer almost overlapped, indicating that the contribution of RIN transfer was negligible under this condition. In contrast, the calculated SNRs with only pump-induced XPM and with both impairments also almost overlapped, indicating that pump-induced XPM was the more significant contribution under this condition. Fig. 6(b) shows the same comparison for the 1480-nm cPUMP, for which pump-induced XPM was larger because the pump wavelength was closer to the signal wavelength. In contrast, pump-induced XPM was smaller for the iPUMP in Fig. 6(c) because of its lower power

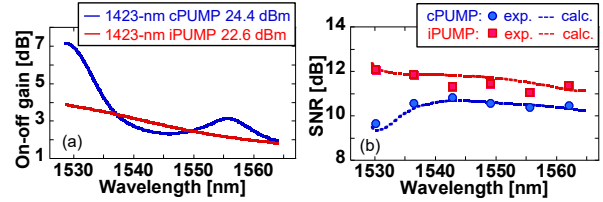


Fig. 7: Evaluation of C-band WDM transmission, (a) on-off forward Raman gain, (b) measured and calculated SNR after 1280-km transmission at 3 dBm/ch signal launch power.

spectral density (PSD) as shown in Fig. 3(b).

The proposed numerical framework was also validated in C-band WDM transmission. Fig. 7(a) shows the on-off forward Raman gain for the cPUMP and iPUMP at 1423 nm under their maximum pump output-power and 3 dBm/ch signal-launch power conditions. Fig. 7(b) shows the corresponding SNR after 1280-km transmission. In the calculation, the same η and μ values as those used in the single-channel case were applied. The reasonable agreement between the experiment and calculation again supports the validity of the proposed framework.

Conclusion

We proposed and experimentally validated a numerical framework for QoT estimation in forward-Raman-amplified coherent transmission. The results indicate that, although pump-to-signal RIN transfer has been regarded as a primary concern in FDRA, pump-induced XPM has a greater impact on QoT than pump-to-signal RIN transfer under the investigated conditions.

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